

V6.2 · Recoverability and Confusion Analysis

A 17 × 17 matrix, and a spectroscopic reason for every off-diagonal cell

Objective	Replace a scalar accuracy with a full confusion structure, and explain each confusion by spectral and component overlap.
Method	For every labelled analyte, the expected motif (from exemplar membership) is compared with the arg-max motif. Each off-diagonal cell is annotated with the cosine between the two motifs' implied spectra, their component-support overlap and their shared bands.
Frozen	Atlas 09ed804a40836f4a05a91ba10900cded; the V6 MSS map M is an input, never rewritten.
Reproduce	results/v6_rebuild/code/p10 → p11 → p12 → p13

Part 7 — recoverability matrix and its spectroscopic explanation

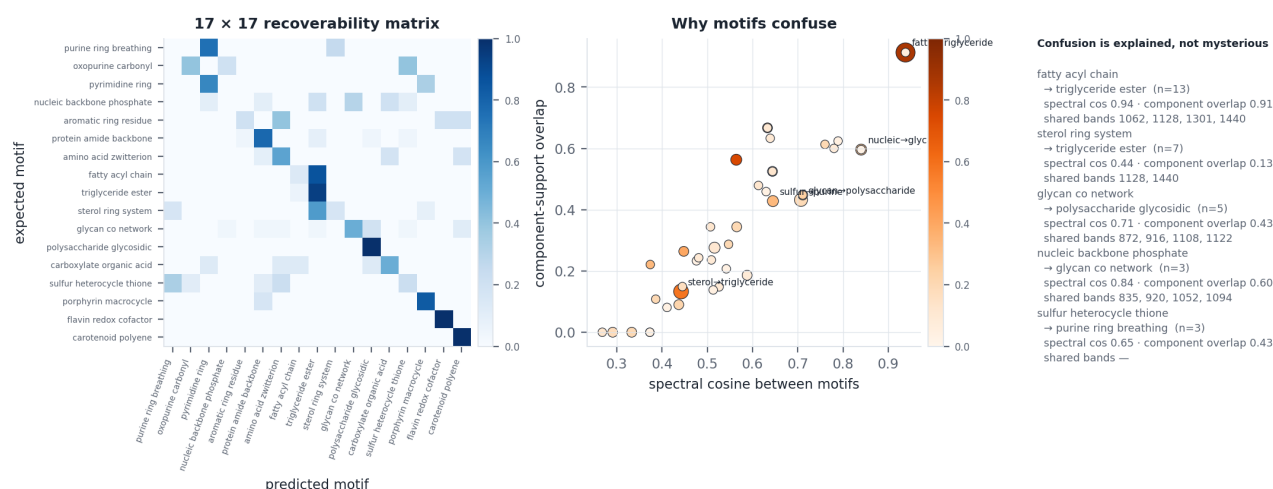


Figure 1 — The recoverability matrix, the confusion drivers, and the top cases.

Results

motif-level top-1	0.503
analytes scored	163
weakest motifs	purine ring breathing, nucleic backbone phosphate, fatty acyl chain, sterol ring system, aromatic ring residue

expected	predicted	n	spectral cos	component overlap	shared bands
fatty acyl chain	triglyceride ester	13	0.938	0.913	1062, 1128, 1301, 1440
sterol ring system	triglyceride ester	7	0.442	0.133	1128, 1440
glycan co network	polysaccharide glycosidic	5	0.707	0.432	872, 916, 1108, 1122
nucleic backbone phosphate	glycan co network	3	0.840	0.596	835, 920, 1052, 1094
sulfur heterocycle thione	purine ring breathing	3	0.645	0.428	—

expected	predicted	n	spectral cos	component overlap	shared bands
glycan co network	carotenoid polyene	3	0.516	0.276	—
purine ring breathing	pyrimidine ring	3	0.564	0.563	1250
nucleic backbone phosphate	triglyceride ester	2	0.333	0.000	1052

The confusions are self-explaining. The single largest is *fatty_acyl_chain* → *triglyceride_ester* (n=13) at a spectral cosine of **0.938** and component overlap **0.913** — the two motifs share four bands (1062, 1128, 1301, 1440) and differ only by the ester carbonyl. That is not a model failure; it is what a free fatty acid and its triglyceride look like. V6 merged them into one theme for exactly this reason, and V6.2's soft layer keeps them adjacent rather than pretending they are separable.

Interpretation

Off-diagonal mass concentrates where spectral cosine exceeds ~0.7. The exceptions are informative: *sterol_ring_system* → *triglyceride_ester* (n=7) has a spectral cosine of only 0.44 and a component overlap of 0.13, so that confusion is **not** explained by motif similarity — it is the known failure of the atlas to isolate the steroid ring system, surfacing again at a third level of the hierarchy.

Limitations

- The expected-motif label is a curated evaluation overlay from exemplar membership, not ground truth.
- The matrix is computed on analyte means, so within-analyte variability is not represented.
- n is small for several motifs (carotenoid 2, polysaccharide 5).

Recommendations

- Treat spectral cosine > 0.9 pairs as a single reporting unit rather than trying to separate them.
- Investigate the sterol confusion at the component level; it is not a motif problem.

Atlas 09ed804a40836f4a05a91ba10900cded verified unchanged. All values recomputed from results/v6_rebuild/.